PTC Quality Control Test Setup & Quality Control Procedure

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v1.4	2026-09-30	C. Zuschmidt	Integrated procedure for automated tests; testing redundancies not yet removed until further finalization

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1 Introduction

This document will guide the reader through the process of testing the Power and Timing Cards (PTCs) used in the DUNE experiment. PTCs sit outside the cryostat and operate at ambient temperature. There is one PTC per warm electronics interface crate (WEIC) which powers up to 6 warm interface boards (WIBs) in the WEIC with 12V at up to 6A. Each DC-DC converter to a WIB can be turned on an off independently. The PTC fans out the encoded timing stream from its front panel SFP to each WIB, and accepts timing transmission back from one WIB at a time via a priority encoding scheme. PTC can monitor its own voltages, currents, and some PCB surface temperatures. It also has an I2C interface to WIBs for sensor reading or arbitrary communication with the FPGA. The PTC has an EtherCAT interface to the detector safety system (DSS), and a gigabit Ethernet (GbE) interface to the slow control (SC) system.

The procedure includes three main stages:

- Reception check: photo taking, visual inspection, vendor documentation check
- QC testing: determine that components are installed correctly by running functional tests for power up, and FPGA and microcontroller programming
- Installation of front panels and heatsinks, and storage for shipping

The document is structured as follows:

- Section 2 provides a detailed overview of all equipment used during QC testing: hardware, software, and tools
- Section 3 offers a quick view of the workflow of each of the processes in the QC testing
- Section 4 shows the step by step procedures required to perform the QC tests

We strongly recommend that the reader goes through this manual entirely before starting to perform any tests.

2 List of Equipment

Tables 1-3 describe all the equipment necessary to perform the QC tests.

2.1 Visual Inspection

Code	Name	Notes
101	Digital camera capable of 12MP	For taking high resolution photographs of both sides of assembled boards
102	Microscope and camera	For reception checkout of defects in front panels and bare PCBs
103	Code Scanner	Tera 1D & 2D Scanner with base and bluetooth dongle, for entering assembled PCBs into DUNE HWDB

Table 1: List of visual inspection components and their corresponding codes



Figure 1: Visual inspection components (microscope and camera)



Figure 2: Visual inspection components (scanner)

2.2 QC Board Test Station

The QC board is the main test structure which PTC plugs into. All the associated hardware and software are listed in Table 2, and illustrated in Figures 3 – Figures 4.

Code	Name	Notes
201	QC test board	Main test structure, with strain relief structure for cabling
202	24V @ 3A laboratory power supply and cable	24V for QC board, 12V for cooling fans
203	48V @ 12.5A power supply	For PTC device under test (DUT)
204	Oscilloscope, at least 2 channels and probes, at least 100MHz bandwidth	For probing QC board test points (and PTC test points if required for debugging)
205	Digital volt meter (DVM)	For probing test points
206	Linux machine	For test scripts, timing server, HWDB
207	3x Ethernet cables	For QC board, PTC GbE connection, and EtherCAT connection
208	GbE media converter	10GTek A7S2-33-1GX1GT-SFP/GT3 for PTC front panel to network switch
209	GbE network switch	For connecting PTC and QC board to Linux machine
210	Windows machine	For running Beckhoff TwinCAT (EtherCAT software)
211	10/100 media converter	tp-link MC100CM Fast Ethernet Media Converter, for connecting PTC front panel to Windows machine
212	3x SFP modules	10Gtek ASF85-24-X2-D (GbE), SFP-100FX-31-I, FS P/N 12713 (EtherCAT), SFP1G-SX-85, FS P/N 11774 (timing)
213	3x fibers	1m LC to LC OM1, FS P/N 42095 (GbE and timing), 1m LC to SC OM3, FS P/N 41751 (EtherCAT)
214	SD card writer	For preparing bootable SD cards for PTC
215	Microcontroller programming pod	Infineon XMD Link, for XMC-4300

Table 2: List of QC test stand components and their corresponding codes

2.3 Personal Protective Equipment (PPE) for Tester

The list of PPE that the tester should wear while performing QC tests is shown in Table 3, and illustrated in Figure 5.

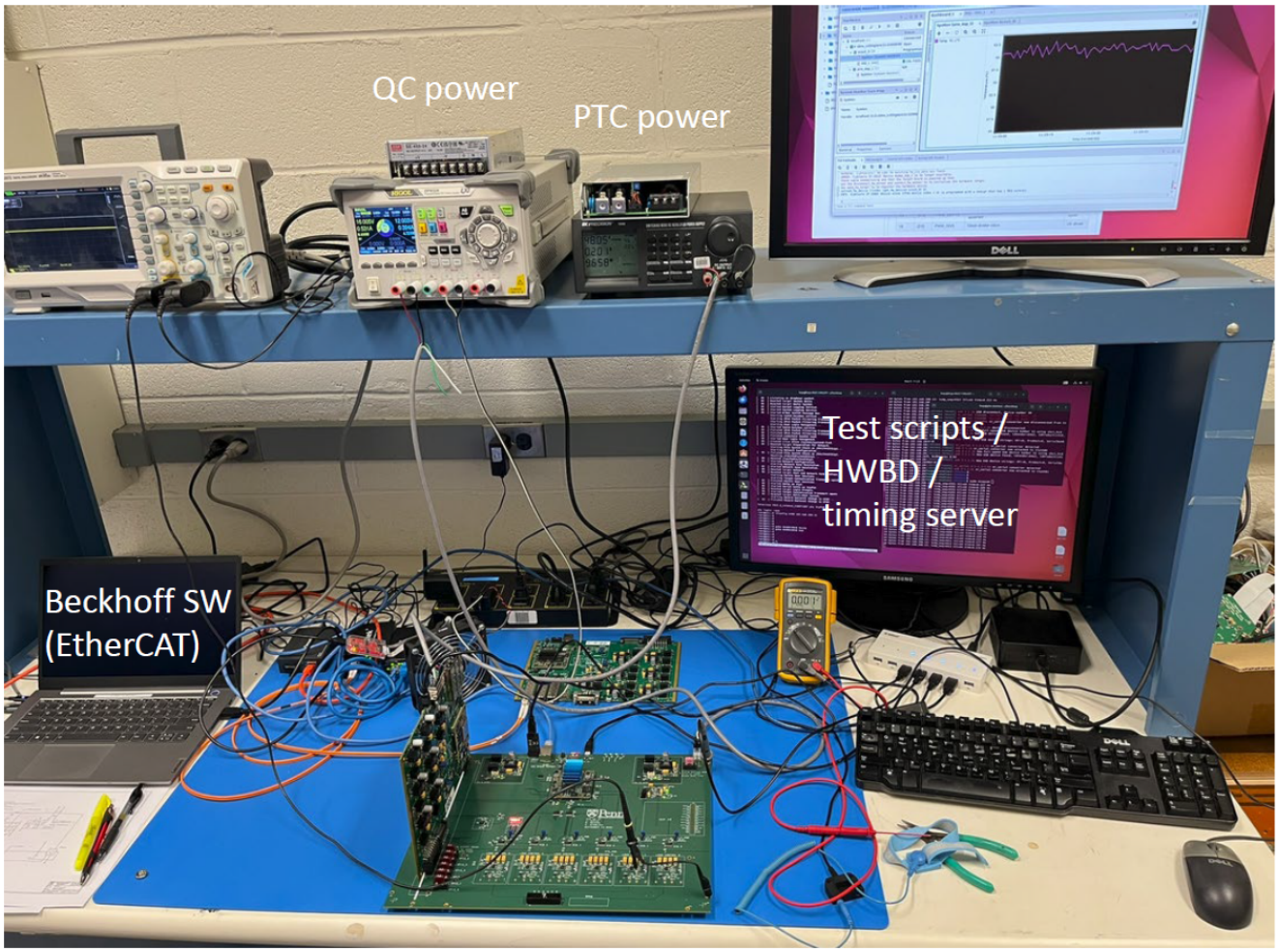


Figure 3: PTC test setup

3 Quick Overview of the Operation Procedure

The overall operation procedure consists of 4 phases: reception checks, manual QC tests, semi-automated QC tests, and final assembly. The reception checks consist of examining front panels and bare PCBs before components are assembled onto the PTCs. After the components are installed and delivered from the assembler, photos are taken and polarized component orientations are checked. A high level overview of these steps are in EDMS document 338406. Manual QC tests consist of measurements that must be taken with DVM or oscilloscope, and tests that involve asserting DIP switches or jumpers on the boards. Semi-automated tests are run by script and generate text output. Final assembly involved installing heatsinks and front panels, and preparing the boards for packaging and shipment.



Figure 4: Closeup of QC board and DUT

4 Itemized QC Procedure

4.1 Reception Checkout

Note: For all visual inspections, the **Acceptance criteria: PASS/FAIL**.

4.1.1 Visual Inspection of Front Panels

An example front panel is shown in Figure 6. The orders of front panels should be examined as follows:

1. **Identification and Counting** – Verify that the correct number of panels has been received as per the purchase order.

2. **Visual Inspection** – A sample (10%) of panels should be examined under the microscope for cleanliness and any signs of physical damage, scratches, or any other defects that could impact functionality. Verify that all required components (e.g., thumb screws) are included.

Code	Name	Notes
301	Long Pants	
302	Long Sleeve Top	
303	Closed Toe Shoes	
304	ESD-safe gloves or Latex gloves	While plugging/unplugging newly assembled PTCs into QC board
305	Antistatic Wrist Strap	Connect to ESD monitoring while handling hardware

Table 3: List of PPE for the tester to wear during QC tests.

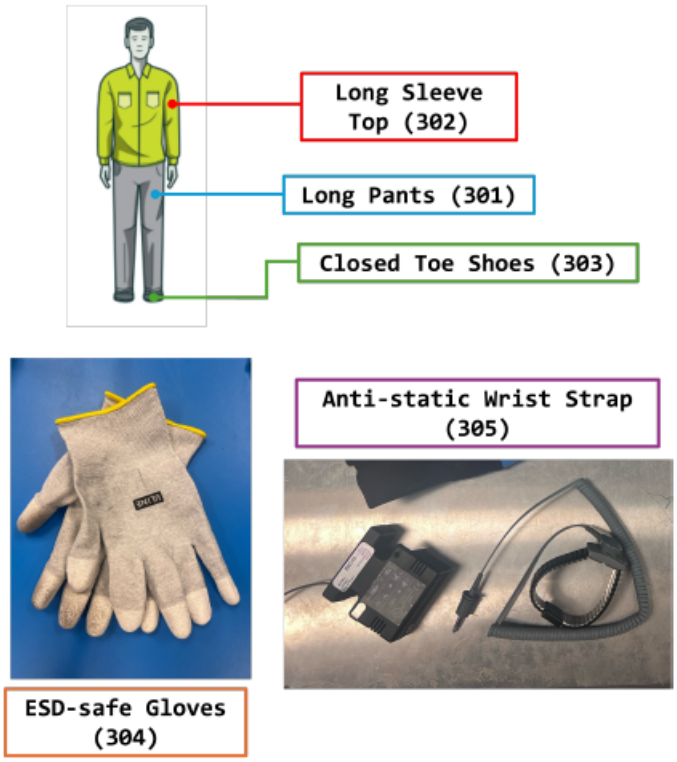


Figure 5: PPE Components

3. **Mechanical fit check** – A sample (10%) of panels should be used to verify overall dimensions, as well as the positions and sizes of mounting holes and connector openings, against the official drawings.

The front panels are not tracked in the DUNE HWDB and not serialized. Any non-conforming panels shall be segregated into a separate box labeled "Rejected front panels" and reported to a UPenn electronics engineer. The non-conformance issues will be reported to the vendor.

4.1.2 Visual Inspection of Bare PCBs

An example bare PCB is shown in Figure 7. The PCBs will arrive from the vendor packaged in ESD safe bags. Do not open the bags for the production PCBs. Sample boards will arrive separately packaged. These samples boards should be removed from their plastic bags to be examined.

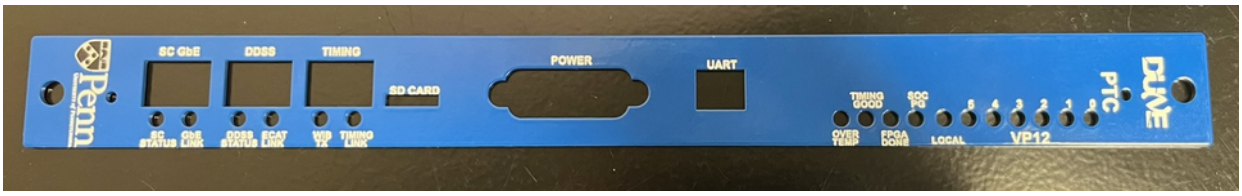


Figure 6: Example PTC front panel

1. **Identification and Counting** – Verify that the correct number of PCBs has been received as per the purchase order.

2. **Reviewing quality certificates**

   (a) Examine the Quality Certificate and QC test reports provided by the vendor and fill the designated form on GoogleDoc. Document any issues in the "Notes" section in the form. If any problems are identified, notify the QC representative.

   (b) Download the completed form from GoogleDoc as Spreadsheet in Microsoft Excel and PDF formats and upload both files to DUNE HWDB.

   (c) Make a copy of QC certificates and test reports and upload them to DUNE HWDB.

3. **Visual Inspection of Sample PCBs**

   (a) Inspect the Sample Boards under the microscope according to the check list provided on GoogleDoc and fill the form. Use the provided reference PCB to make a judgment on various check items. Record any problems in the "Notes" section in the form. If defects are found, notify the QC representative. The checklist is reproduced here for clarity:

      i. Overall PCB appearance and cleanliness
      ii. Surface quality: existence of pit, scratch or pin holes on printing traces or pad
      iii. Inspect PCB edges for possible defects
      iv. Inspect both sides of PCB
      v. Check for any other noticeable defects, damages, irregularities, etc.

   (b) Download the completed form from GoogleDoc as Spreadsheet in Microsoft Excel and PDF formats and upload both files to DUNE HWDB.

   (c) Attach a sticker to the Sample Board with the following information:

      i. PCB serial number
      ii. PCB production batch number
      iii. Inspection date
      iv. Initials and last name of the person who performed the inspection

The PTCs are not tracked by serial number until the assembly process is complete. Any non-conforming sample bare PCBs at this stage shall be segregated into a separate box labeled "Rejected bare PCBs" and reported to a UPenn electronics engineer. The non-conformance issues will be reported to the vendor.

4.1.3 Visual Inspection of Assembled PCBs

An example of an assembled PCB is shown in Figure 8. The PCBs will arrive from the vendor packaged in ESD safe bags. At this stage, all PPE should be worn to prevent damage from ESD discharge and contamination.

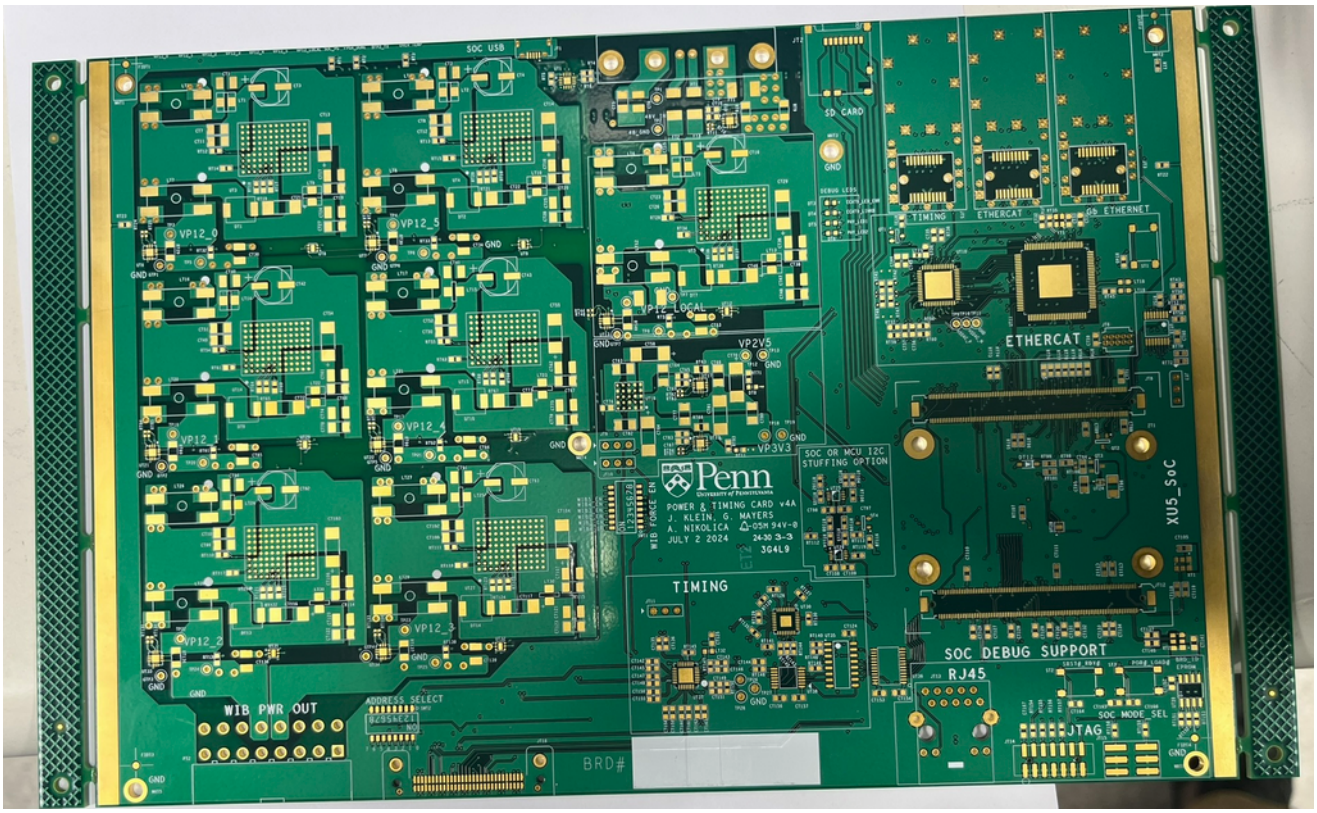


Figure 7: Example bare PCB

1. **Identification and Counting** – Verify that the correct number of PCBs has been received as per the purchase order.

2. **Reviewing quality certificates**

   (a) Examine the Quality Certificate and QC test reports provided by the vendor and fill the designated form on GoogleDoc. Document any issues in the "Notes" section in the form. If any problems are identified, notify the QC representative.

   (b) Download the completed form from GoogleDoc as Spreadsheet in Microsoft Excel and PDF formats and upload both files to DUNE HWDB.

   (c) Make a copy of QC certificates and test reports and upload them to DUNE HWDB.

3. **Visual Inspection of Sample PCBs**

   (a) Take a high resolution photograph of the front and back of the boards, and upload to the DUNE HWDB.

   (b) Inspect each assembled PCB under the microscope according to the check list provided on GoogleDoc and fill the form. Use the provided reference PCB to make a judgment on various check items. Record any problems in the "Notes" section in the form. If defects are found, notify the QC representative. The checklist is reporduced here for clarity:

      i. Missing solder joints, particularly on through-hole components that also have surface-mount (SMD) pins

      ii. Excess solder paste or bridging between adjacent pins

      iii. Unintended components left on the board (e.g., from pick-and-place errors)

      iv. Residues or contamination due to insufficient cleaning during assembly

v. Missing components (DDR memory, jumpers, etc.)

vi. Incorrect orientation of ICs and polarized components (check pin 1 marker on ICs, and [+] marker on tantalum and polymer capacitors)

(c) Document any observed issues and include a written description of the defect as well as a closeup photograph of the defect.

(d) Tag any non-conforming boards as "Visual Inspection Failed" and segregate them into a box labeled "Rejected Assembled PCBs".

(e) All boards, whether pass or fail, will have their serial codes scanned into the DUNE HWDB, and the result of the pass or fail will be noted.

(f) Download the completed form from GoogleDoc as Spreadsheet in Microsoft Excel and PDF formats and upload both files to DUNE HWDB.

Any bare PCBs rejected from the visual inspection process and placed in the box labeled "Rejected bare PCBs" shall be reported to a UPenn electronics engineer. The non-conformance issues will be reported to the vendor.

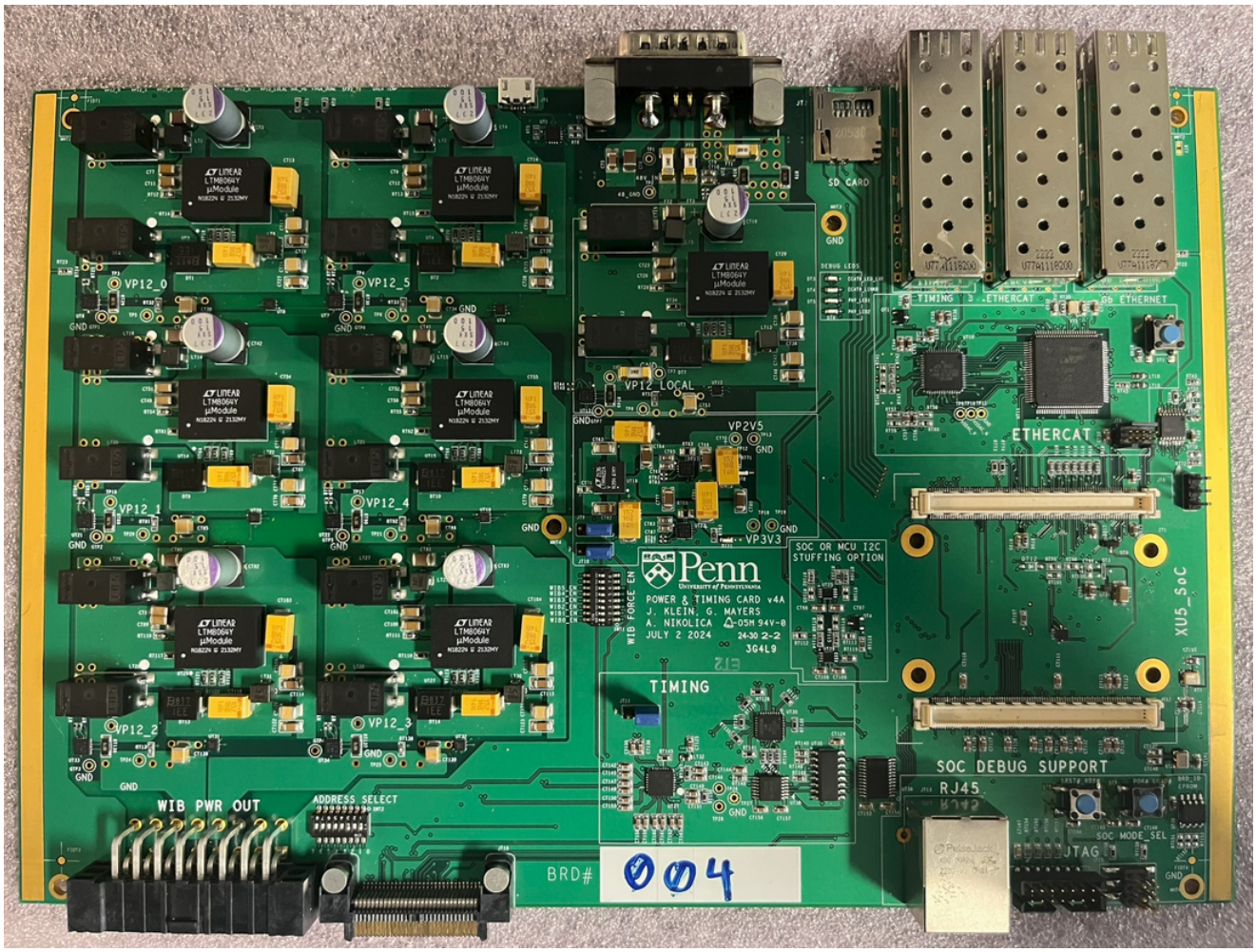


Figure 8: Example assembled PCB

4.2 QC Test Stand Preparation

4.2.1 `dune-daq` Setup

The software needed for `dune-daq` is here. Refer to the installation guide here.

1. Run the following sequence of commands. **Note:** The # symbol indicates an inline comment and is not part of the command sequence.

```
# the commands below need to be done every time the machine is rebooted
bash # invoke bash to install
cd [your setup directory]
source /cvmfs/dunedaq.opensciencegrid.org/setup_dunedaq.sh
setup_dbt dunedaq-v3.1.1

# the commands below only need to be done once
dbt-create -c -n N22-11-08 dune-daq-timing-dcsk # explicitly setup known
                                                  working version
cd dune-daq-timing-dcsk/sourcecode
git clone https://github.com/DUNE-DAQ/timing.git
cd timing
git checkout tags/"nocdr_integration/b0"
cd ../../

# The commands below need to be done every time the machine is rebooted
source dbt-env.sh
dbt-workarea-env
dbt-build
```

2. Create the following file named `local.xml` inside the `dune-daq-timing-dcsk` folder. **Note:** This only needs to be done once.

```
<?xml version="1.0" encoding="UTF-8"?>
<connections>
    <connection id="MST_FMC" uri="ipbusudp-2.0://192.168.200.16:50001"
        address_table=
        "file://sourcecode/timing/config/etc/addrtab/v7xx/master_fmc/top.xml" />
    <connection id="EPT_0" uri="ipbusudp-2.0://192.168.200.16:50001"
        address_table=
        "file://sourcecode/timing/config/etc/addrtab/v7xx/endpoint_fmc/top.xml" />
</connections>
```

4.2.2 Timing Master Setup

Note: The following steps only need to be done once per powerup.

1. Download the file `ouroboros_pc053d_nocdr-integration-b0_sha-01d4e48e_runner-slu9p8x4-project-19909-concurrent-1_220727_1123.tgz` from here. Unzip the file completely (it will create a folder). The file in this folder with extension `.bit` is the file that will be used to program the timing board.

2. Ensure the timing board is powered on (its power brick should be plugged in) and that it's connected to the Xilinx JTAG cable.

3. Open Vivado on the Linux machine, and select Hardware Manager. The JTAG cable should be visible in the panel that appears.

4. Right click on the JTAG cable and select "Auto Connect". A Xilinx part number will appear – this is the timing board's FPGA.

5. Select "Program" and select the file downloaded in the previous step. Program the device.

6. Connect to the serial UART with `minicom -D /dev/ttyUSB[x]` where x is probably 0 if there are no other devices connected – you can use `sudo dmesg` to see what the last connected device was. The settings should be: 19,200 baud, 8 bits, no parity bits, 1 stop bit, no hardware flow control. Use the key combo: Ctrl+A, then Z, and O to configure the setttings in `minicom`. Ctrl+A and X exits.

7. `help` shows a list of commands. Confirm the serial port returns output.

8. `write` and then Enter

9. followed by `0xc0a8c810` and Enter at the next prompt will set the IP address as 192.168.200.16.

10. `ping` the board to confirm the setting took effect.

11. The serial port can be disconnected at this point.

12. Now at the Linux command prompt used above to setup `dune-daq`, issue the following sequence of commands. This will set up the timing board to send out a valid timing stream on its SFP.

```
dtsbutler -c local.xml io MST_FMC reset
dtsbutler -c local.xml mst MST_FMC synctime
dtsbutler -c local.xml io EPT_0 reset
dtsbutler -c local.xml mst MST_FMC control-timestamp-broadcast
```

4.2.3 Preparing a DUT for Testing

1. Referring to Figure 3, make sure all cables are plugged in while power is off. Observe that the strain relief is securing the cables drooping from the lab hutch where power supplies are located.

2. Next, slide the first PTC DUT to the QC board as per Figure 4, inserting into the black guardrails, and ensuring that you press the DUT firmly and with even pressure so that the Samtec connectors mate flush with the QC board.

3. Next, power on the QC board:

   (a) Confirm that the latest version of QC board firmware (not PTC firmware) is loaded on the SD card plugged into the QC board.

   (b) Power on the QC board by turning on the 24V lab supply. Initial current consumption should be about 400mA. After a few seconds, the amber LED on the Enclustra board will start blinking at a few Hz and remain blinking. If the amber LED is solid on, or fails to come on, contact a UPenn electronics engineer for assistance.

(c) Establish an Ethernet connection with the QC board. It should come up as 192.168.200.10 on the network. You can ssh into the QC board as user root. **Note:** if the Ethernet connection fails to link, the serial UART can be used to diagnose. It will come up as /dev/ttyUSB[x], where x is typically 0 or 1 (since there are two serial interfaces in the system – PTC and QC board). The settings should be: 115,200 baud, 8 bits, no parity bits, 1 stop bit, no hardware flow control. Consult a UPenn electronics engineer for assistance if the link fails.

(d) Turn on the 12V to the QC board heatsink fans (this is CH2 on the lab supply).

4. Next, prepare the bootable SD card for the DUT (these steps are listed on the github page for the PTC firmware, but are repeated here for clarity):

   (a) There will be a standard reference image for the firmware. The image consists of 4 files, explained below.

   (b) Ensure that the networking-stack package is enabled for later ethernet testing

   (c) Format the SD card by using `sudo gparted` to create two partitions on an SDHC micro-SD card: a 256MB FAT32 partition named BOOT, and the remainder of the disk should be an EXT4 partition called ROOT.

   (d) Copy the following files to the BOOT partition: BOOT.bin, image.ub, boot.scr. These files load the FPGA image, and start the initial Linux boos.

   (e) Copy the file rootfs.tar.gz to the ROOT partition. Then `gunzip` and `tar -xvf` the file to uncompress the entire contents. This is the root filesystem.

5. Next, insert the SD card prepared in the previous step into the PTC's SD card slot. Ensure the SD card clicks firmly into place with little force. Use tweezers if necessary. If the SD card is not engaging, it may be inserted upside down. Do not force the card.

6. Next, install the jumpers and prep the board:

   (a) As per Figure 8, install the following 3 blue 2-position jumpers:

   (b) Timing jumper JT11: position 2–3

   (c) 3.3V enable jumper JT9: position 2–3

   (d) 2.5V enable jumper JT10: position 2–3

   (e) All other 2 or 3 position headers are left with no installed jumpers

   (f) Remove the amber colored film from the DIP switches labeled SWT1 and SWT2. SWT1 will be used the enable DC-DC converters for these initial tests only, before the SoM is installed. SWT2 is used to set the crate addresses.

7. Finally, scan the DUT serial number with the scanner. **Acceptance criteria: PASS/-FAIL, with serial number noted in HWDB**.

4.3 Manual QC Tests

1. **Power up**

   (a) Turn on the 48V power supply. The PTC without SoM installed should consume less than 100mA. **Acceptance criteria:** $\leq 100mA$.

   (b) Temporarily remove the clear plastic plate on the QC test stand so that probe points can be reached.

   (c) Voltage check: measure with the DVM at the test point on the PTC labeled VP12_LOCAL and ensure it is 12V. **Acceptance criteria: 12.00–12.35V**.

   (d) Ensure that the front panel LED labeled 12V_LOCAL is on. **Acceptance criteria: PASS/FAIL**.

4.3.1 DC-DC Converter Checks

1. **Voltage Check**

   (a) Ensure the Enclustra SoM is NOT installed on PTC board at this point.

   (b) Using tweezers or a small flat head screwdriver, set SWT1 in the first position (that is: 8b 0000 0001).

   (c) On the QC board, measure with the DVM at the test point labeled RIPPLE_0 and ensure it is 12V. **Acceptance criteria: 12.00–12.35V**.

2. **Switching Frequency** With the oscilloscope attached to the probe point, set to AC coupling, 20mV scale, and 2us timescale, observe the period of noise as shown in Figure 9. **Note:** Do not rely on the oscilloscope frequency measurement function, as it is likely to be confused by the low resolution of the pulses. **Acceptance criteria: period 1.0–1.1μs (frequency 917kHz – 1MHz)**.

3. **Front Panel LEDs** Observe that the LED labeled WIB_0 is illuminated. **Acceptance criteria: PASS/FAIL**.

4. **Repeat** with each of the other 5 DC-DC converters. So, for instance, the second converter will be set to 8b 0000 0010 on SWT1, and RIPPLE_1 will be measured, and the front panel LED labeled WIB_1 will be observed. Acceptance criteria are the same in all cases.

5. **IMPORTANT:** All DIP switches must be left in the OFF position before moving on to the SoM tests.

6. Power off PTC, disconnect all cables.

7. Remove PTC from the QC board test stand by inserting Allen wrenches into the holes on the support structure on each side, and using the wrenches as levers to lift PTC up and away from the QC test stand.

8. Re-install the clear plastic plate on the QC test stand for the remaining tests.

4.3.2 SoM Tests

1. **Install Test**

   (a) Scan the Enclustra module QR code, and enter this to be tracked in the database.

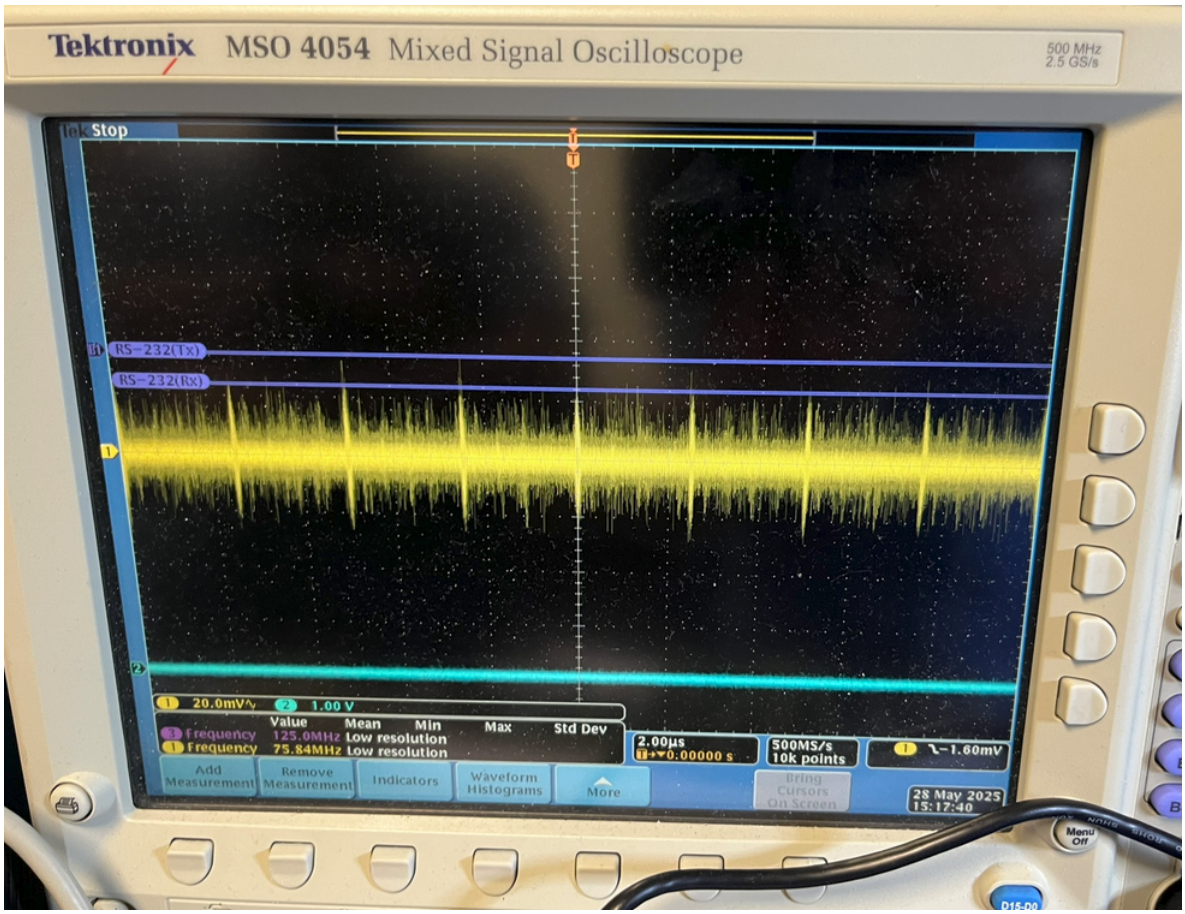


Figure 9: DC-DC converter switching waveform

(b) Ensure that the Enclustra SoM can be inserted into connector with no excessive force. Do not force the connectors to mate to avoid bending connector pins or damaging the SoM. **Acceptance criteria: PASS/FAIL, with the serial number noted in HWDB**.

(c) **IMPORTANT: observe the polarity of the Enclustra SoM when inserting! There are two sets of unequally spaced holes on each side of the SoM that must line up with the corresponding holes in the PTC.**

(d) After the SoM is installed, re-insert the DUT into the QC test stand.

2. **Boot Test**

(a) Turn on the 48V power supply. Initially, PTC will consume about 100-200mA. After a few seconds, the amber LED on the Enclustra board will start blinking at a few Hz and remain blinking. If the amber LED is solid on, or fails to come on, contact a UPenn electronics engineer for assistance.

(b) Ensure that the front panel LED labeled 12V_LOCAL is illuminated. **Acceptance criteria: PASS/FAIL**.

(c) Ensure front panel PG LED is illuminated. **Acceptance criteria: PASS/FAIL**.

(d) Ensure front panel FPGA DONE LED is illuminated. **Acceptance criteria: PASS/FAIL**.

(e) Ensure the serial port can be connected to. It will come up as /dev/ttyUSB[x], where x is typically 0 or 1 (since there are two serial interfaces in the system – PTC

and QC board). The settings should be: 115,200 baud, 8 bits, no parity bits, 1 stop bit, no hardware flow control. The login name is root. **Acceptance criteria: PASS/FAIL**.

3. **Repeat Boot** Issue the command `init 0` at the DUT serial port and wait for the prompt to return a message that it has stopped all services. Turn power off for 60 seconds, and then back on again. Repeat the bootup procedure 5 times. **Acceptance criteria: PASS/FAIL**.

4.4 Semi-automated QC Tests

4.4.1 DC-DC Converter Load Test

1. **Setup** Boot up the DUT again while it is plugged into the powered-on QC board. Now commands will be issued on the DUT and QC board to exercise different loads.

2. **Turn on PTC** While ssh'd into the DUT, type `python3 power_on_wib.py 0 on` for the first DC-DC converter.

3. **Enable PWM driver** While ssh'd into the QC board, type `poke 0x8002004c 0x316` followed by the value `0x216`. This sets the PWM driver to 2MHz and 25% duty cycle. It can be tested by measuring the PWM probe point.

4. **Measure current draw** The DUT should be drawing approximately 1.5A. Confirm on the 48V power supply that the current consumption has increased by 500mA (Note that the current consumption of 2A at 12V is equivalent to 500mA at 48V. There is some inefficiency in the converter – the load side current will be measured more accurately with I2C sensors in the next section.) **Acceptance criteria:** $1.5A \pm 100mA$.

5. **Repeat with other load values** The values of `0x226, 0x236, 0x23e` will test 50%, 75%, and approximately 100% load on the converter. Do not leave tests over 50% running for more than a few minutes. Confirm the coresponding current draws are 3A, 4.5A, and 6A. **Acceptance criteria:** $3A \pm 100mA$, $4.5A \pm 100mA$, $6A \pm 100mA$.

6. **Measure voltage ripple** At the highest load, measure the test point RIPPLE_0 with the scope set on 200mV scale, 200ns period, and AC coupling, and note the AC ripple with an RMS measurement. **Acceptance criteria:** $\pm 200mV$. An example is shown in Figure 10.

7. **Repeat with other converters** The addresses of `0x80020048, 0x80020044, 0x80020040, 0x8002003c, 0x80020038` correspond to converters 1 through 5. Repeat the test for all converters. Acceptance criteria are same as above.

4.5 Automated Test Setup

1. Ensure the Linux PC is connected to the DUT via USB, and is listed as /dev/ttyUSB0. Confirm physical ethernet link.

2. Open 2 terminals on the Linux test machine and cd into **PTC-Firmware/board_qc directory** in each

3. In one terminal, run:

   (a) `python3 -m venv .env`

   (b) `source .env/bin/activate`

   (c) `pip install -r ptctestclient/requirements.txt -r ptctestsuite/requirements.txt`

4. In the test suite terminal, run:

   (a) `source .env/bin/activate`

   (b) `python -m ptctestsuite`



Figure 10: Example of ripple measurement (CH2, blue)

5. **Confirm network parameters** in ptctestsuite/config/parameters.py match the bench: PTC at 192.168.20.100, WebSocket port 8765

6. **Authenticate to the HWDB.** On launch, the suite runs htgettoken against htvault-prod.fnal.gov. Follow the printed instructions to log in with your Fermilab Services account.

7. Note: **Board network bring-up.** The suite opens the serial port (/dev/ttyUSB0), logs into the PTC, and assigns eth1 its IP. Note: if it reports a failure on the default port, enter an alternate (e.g. /dev/ttyUSB1) when prompted, but confirm it's PTC serial. If it fails again it logs PTC Network Configuration Failed! and aborts the session.

8. **Client launch**. The suite SSHes into the DUT and starts ptctestclient, then waits up to 20 seconds for the client's Server Listening message. Note: if the client does not report ready in time, the session logs a critical error and the WebSocket connection will fail. **Acceptance criteria: [PASS/FAIL]**.

9. **Automated tests run.** The suite connects to ws://192.168.20.100:8765, printing Running ¡test_names¿ with parameters. Each result is recorded as **PASS / FAIL / ERROR** against the serial number.

   (a) Automated test suite includes: IV sensors on PTC, IV sensors on QC board, temperature sensing, EEPROM, memory

10. Manual tests. The suite prompts the tester with a (y/n) question and, where the test accepts a reading, a value prompt. Enter observed result and measurement. **PASS/FAIL per test**

(a) Manual tests include ethercat, GbE, [TBD]

11. **Notes.** Answer the notes prompt; if yes, a full-screen text box opens for shift comments

12. **Confirm upload** The suite writes the record to the HWDB in three POSTs: create the component entry (returns a part ID), post the test datasheet, then capture a webcam image of the board and post it. Confirm the log reports each upload succeeded. **[PASS/FAIL]**

4.5.1 I2C Sensor Tests

1. **Setup** The test script will simultaneously test all sensors in the list below, as well at the EtherCAT link in the QC tests below.

   (a) On the DUT, turn all DC-DC converters on with `python3 power_on_wib.py [x] on` where the command is repeated 6 times with [x] being 0–5.

   (b) Then, on the DUT, run `python3 ecat_test_1b.py [filename].txt` where filename is the logfile; choose any name (for example, the serial number of the DUT).

   (c) A sample output is shown in Figure 11. The full set of values to be written into the HWDB will be parsed by a Python scope (to be written) and assigned PASS/FAIL values based on the acceptance criteria.

2. **Main DC-DC Converter Sensors**

   (a) Voltage. Each 12V output to WIB is read from the sensor. **Acceptance criteria: 12.00 – 12.35V**.

   (b) Current. Measured at 25% load. The purpose of this test is to ensure the current measurement resistor is installed and the correct value, so a comprehensive tests at different loads does not need to be repeated. **Acceptance criteria:** $1.5A \pm 100mA$.

3. **SoM Sensor**

   (a) Voltage. **Acceptance criteria: 12.00 – 12.35V**.

   (b) Current. **Acceptance criteria:** $\leq 2A$.

4. **Auxiliary Sensors**

   (a) 2.5V Local Voltage. **Acceptance criteria: 2.4 – 2.6V**.

   (b) 2.5V Local Current. **Acceptance criteria:** $\leq 2A$.

   (c) 3.3V Local Voltage. **Acceptance criteria: 3.2 – 3.4V**.

   (d) 3.3V Local Current. **Acceptance criteria:** $\leq 2A$.

5. **Temperature Sensors**. All 8 temperature sensors should read close to ambient temperature. Since there are variations in air flow, local board temperature based on how long a DC-DC converter was recently switched on or off, the acceptance criteria is not strict. The purpose of the test is to ensure that the temperature sensors are installed. **Acceptance criteria: Ambient temperature** $\pm 5C$.

6. **Sensor Extended Functionality**

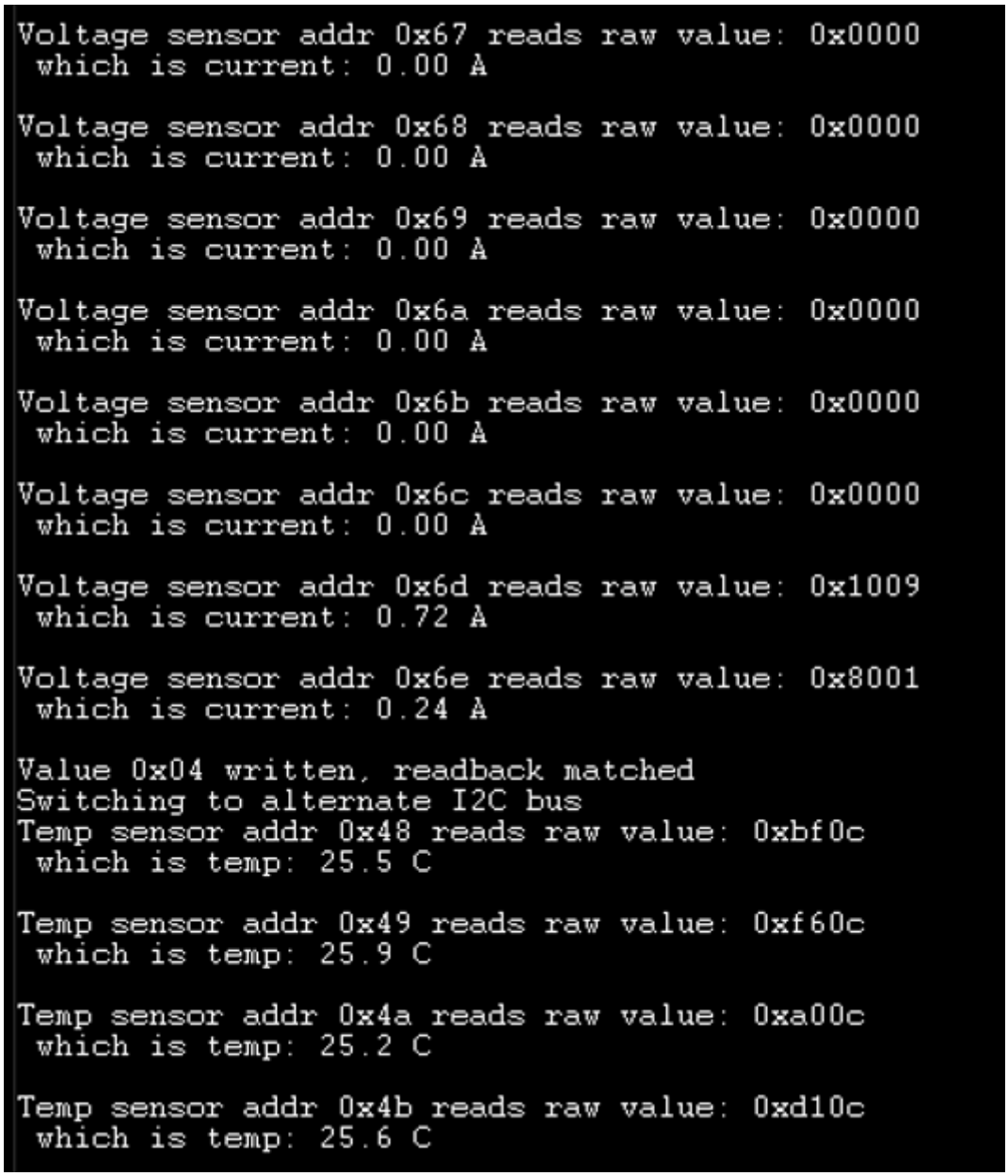


Figure 11: A sample selection of I2C output. Note: no load applied.

(a) ALARM. From the QC_DEV branch of the github repository, run the `qc_firmware\scripts\test_scripts\qc_board_tests\alert_assert.py` script. This will go through each sensor, program a threshold such that the ALERT pin is asserted, and readback the ALERT register over I2C. **Acceptance criteria: PASS/FAIL**.

(b) Front panel OVER TEMP LED. For any of the above ALARM line asserts, the OVER TEMP LED should illuminate in RED on the front panel. **Acceptance criteria: PASS/FAIL**.

4.5.2 Addressing

1. **Crate Address DIP Switch**.

   (a) Temporarily remove the clear plastic plate on the QC test stand.

(b) For each of the 8 DIP switches, switch to ON position with tweezers or a small flathead screwdriver.

(c) Ensure the corresponding crate address ID is illuminated. **Acceptance criteria: PASS/FAIL**.

(d) Re-install the clear plastic plate on the test stand.

2. **EEPROM R/W Test**. From the QC_DEV branch of the github repository, run the `qc_firmware\scripts\test_scripts\eeprom.py` script. This will read the globally unique serial ID from the EEROM. **Acceptance criteria: PASS/FAIL, with serial number noted in HWDB**.

4.5.3 Microcontroller Tests

1. **Setup** The complete setup instructions are shown on the PTC github page README, and are repeated here for clarity:

   (a) The EtherCAT microcontroller project is based on the Infineon reference design. Ensure that the same version of tools are installed. Beckhoff TwinCAT3 is needed to initiate a connection to PTC. Infineon DAVE v4.1.4 or higher and the Beckhoff SSC tool v5.12 are used for creating the initial project, but is only needed for development.

   (b) The first time the link is set up, or if any configuration changes, copy the XMC_ESC.xml file (the "ESI file") from `ethercat/xmc4300_proj/xmc4300_proj/SSC/XMC_ESC.xml` to `C:\TwinCAT\3.1\Config\Io\EtherCAT` on the host PC.

   (c) Connect a 100Base-FX SFP to SFP1, with an LC-to-SC fiber connection. For bench testing, this can be connected to a optical-to-fiber converter (like the tp-link MC100CM).

   (d) From the main branch in github, open up the TwinCAT project in the `ethercat/` directory.

   (e) Now program the microcontroller.

2. **Program MCU**

   (a) Programming over JTAG cable. After connecting the Infineon XMC Link cable to J14, click on the "Debug" button (looks like a bug) in the DAVE project opened in the previous step. This will compile and load the project. **Acceptance criteria: PASS/FAIL**.

   (b) In-situ programming using FPGA. After obtaining a shell, run the command `openocd -f ptc-flash.cfg -c "set FIRMWARE { /home/root/xmc4300_firm.elf }`. **Acceptance criteria: PASS/FAIL**.

3. **Test EtherCAT Link**

   (a) Double click on "Device 1 (EtherCAT)" and click on the "Adapter" panel. Ensure that the correct PTC Ethernet interface is selected.

   (b) Right click on Device 1, and click "Scan". A message saying "Configuration matched" should appear.

   (c) Now click on the "Online" tab. PTC should appear as a "Box" and be labeled "XMC_ESC".

(d) In the top menu bar, click on the "Toggle Free-Run State" icon (a red circular arrow).

(e) Front panel LINK LED. **Acceptance criteria: PASS/FAIL**.

(f) TwinCAT link test. Now the PTC can be changed to "Op" mode in the Online panel. There will be constant frame counters in this panel, and no lost frames. This means PTC is maintaining an EtherCAT link. The PTC Debug LEDs will show: ERR=GREEN, LINKB=GREEN, PHY_KED2=BLINK GREEN. **Acceptance criteria: PASS/FAIL**.

(g) Front panel ECAT RUN LED. **Acceptance criteria: PASS/FAIL**.

4.5.4 Timing

Note that the timing system was set up in a previous step, and should be in a state where it is transmitting valid information.

1. **RX Test**. Run the following sequence of commands:

   (a) Timing lock on PTC

   ```
   root@ptc:~# python3 setup_timing.py
   root@ptc:~# poke 0x80020004 0x00010400
   root@ptc:~# peek 0x80020104
   0x00073f00 # timing endpoint reset
   root@ptc:~# poke 0x80020004 0x00010000
   root@ptc:~# peek 0x80020104
   0x18073f00 # endpoint out of reset, and showing 0x8 status (good)
   ```

   Acceptance criteria: PASS/FAIL.

   (b) Front panel LINK LED. **Acceptance criteria: PASS/FAIL**.

   (c) Front panel RX LED. **Acceptance criteria: PASS/FAIL**.

   (d) Timing lock on QC board (NOTE: Only for ch0; add steps for ch1-5.)

   ```
   root@qc:~# poke 0x80020004 0x00000400
   root@qc:~# poke 0x80020004 0x00000000
   root@qc:~# peek 0x80020104
   0x18000000 # endpoint out of reset, and showing 0x8 status (good)
   ```

   Acceptance criteria: PASS/FAIL.

2. **TX Test**

   (a) To measure round trip time, run the following sequence of commands:

   ```
   dtsbutler -c local.xml mst MST_FMC align measure-delay 0
       # should get ~100 clock ticks
   dtsbutler -c local.xml mst MST_FMC align toggle-tx 0 --on|off
       # should turn LED on
   ```

   The RTT in milliseconds should be displayed. The RTT will vary based on testing conditions, so the purpose of the test is to ensure that the priority encoder and clock multiplexer are installed correctly, and that that the timing stream is routed back through the front panel. **Acceptance criteria: PASS/FAIL**.

   (b) Front panel TX LED. The LED will temporarily illuminate when the RTT test is conducted. **Acceptance criteria: PASS/FAIL**.

4.5.5 Ethernet Tests

1. **Ping Test**

   (a) Front panel LINK LED. **Acceptance criteria: PASS/FAIL**.

   (b) Ping. The DUT should come up as 192.168.200.12 on the network. Issue the `ping` commands from the Linux test machine and ensure the DUT responds. **Acceptance criteria: PASS/FAIL**.

2. **Throughput Test**. Using iPerf3, run the command `iperf3 -s` on the DUT through minicom, followed by `iperf3 -c 192.168.200.12` on the Linux test machine. Alternatively, You can ssh into the PTC as user root. Use the `dd` command to transfer the 1GB file called `thru_test.bin` to `/dev/null`. **Acceptance criteria:** $> 12.5MB/s$.

3. **Front Panel AUX LED**. Illuminate the LED by writing the register address TBD. Note: this is a spare LED that may have a future functionality related to the GbE interface. For now, it is toggled with a register bit. **Acceptance criteria: PASS/FAIL**.

4.6 Mechanical Installations

1. **Heatsink Install**

   (a) Clean the top of the LTM-8064 DC-DC converters with an alcohol swab.

   (b) The thermal interface material (TIM) has double sided adhesive. Remove one side of the adhesive, and apply to the top of each LTM-8064.

   (c) Remove the other side of the adhesive on the TIM and apply the heatsink with even pressure. An example of installed heatsinks is shown in Figure 12.

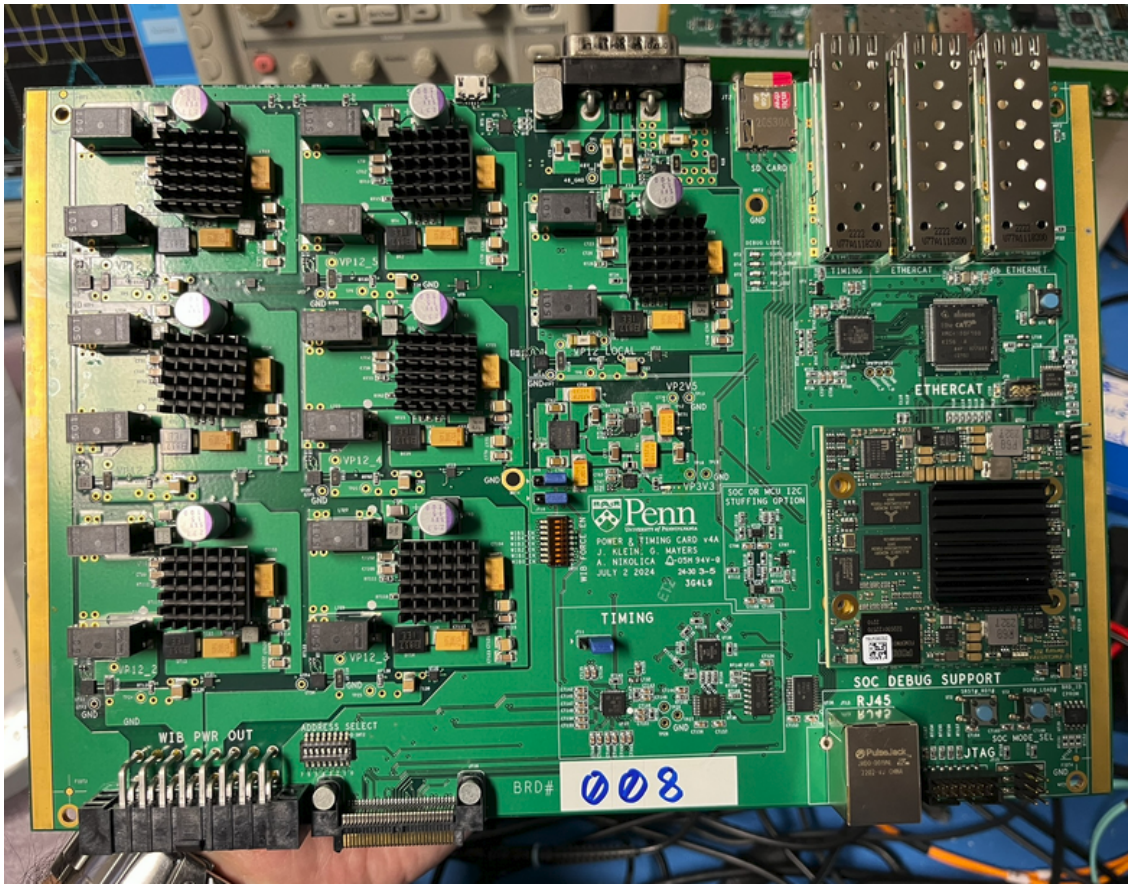


Figure 12: PTC with installed heatsinks

2. **Front Panel Install**

   (a) Insert light pipes into each LED hole and secure with rubber grommet.

   (b) Attach the front panel to the DUT at the 2 L-bracket locations. is shown in Figure 13.

3. **Packing** The PTC is now ready to be packed in an ESD bag and placed in a secure location for shipment. Ensure that the overall test result for the DUT is entered in the HWDB. **Acceptance criteria: PASS/FAIL**.



Figure 13: PTC with installed front panel. Note: light pipes not shown in this picture.

4.6.1 QC Test Stand Tear-down

After testing is finished for a shift, uninstall any DUT still plugged in. On the QC board, issue the command `init 0` and wait for the power-down message (a few seconds). Then power off the 24V to the QC board. Finally, power off the 12V to the QC heatsink fans.

4.7 WEIC Tests of a Sample of PTCs

A subsample of fully tested and assembled PTCs (including heatsinks and front panels) will be sent to BNL for insertion into a WEIC, as in Figure 14. For this test, the sample PTCs will be instrumented with temperature sensors on each of the main 12V DC-DC converters before insertion into the crate. Then 6 WIBs will be powered on via the PTC, and all FEMBs will be turned on and configured for nominal operating conditions. The procedures for this are defined in the WIB QC procedure.

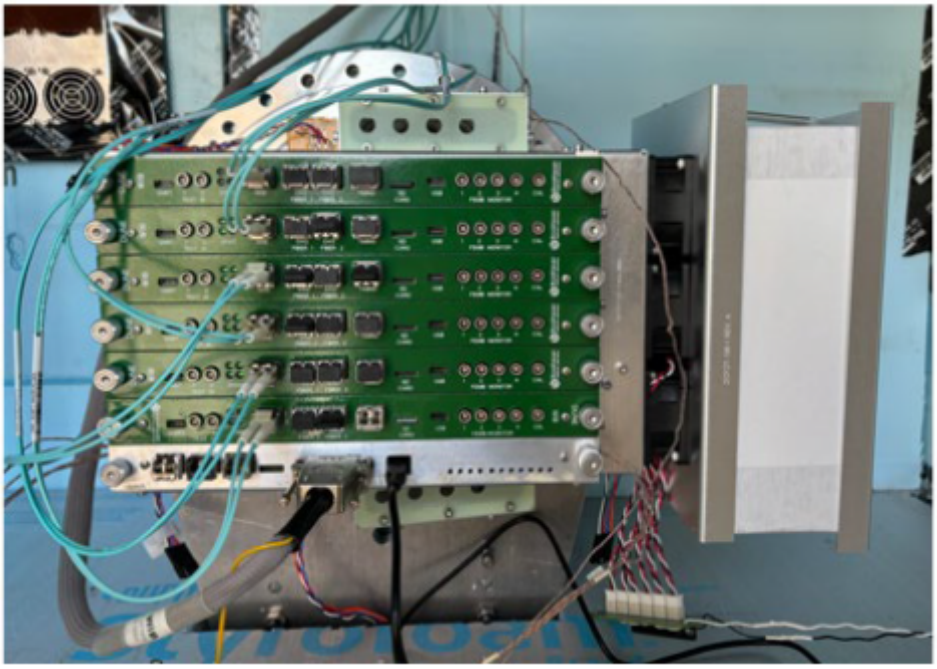


Figure 14: PTC inserted into WEIC

Temperature as measured on top of heatsinks **Acceptance criteria:** $< 60C$ for all heatsinks.

5 End of Shift Checklist

Date:
Time:
Tester's Name:

Tested PTC SN	Result (PASS/FAIL/Other)	Comments

Notes: